

Week 7 — Alternative Formulations, and Branch-and-Bound

OL7014 Integer Programming • Instructor: Prof. H. Park • Week of Oct 5, 2026

1. Alternative formulations — one X , many P

remember from week 5 — the **LP relaxation**:

$$z_{\text{IP}} = \max_x \{c^\top x : Ax \leq b, x \in \mathbb{Z}_+^n\} \leq z_{\text{LPR}} = \max_x \{c^\top x : Ax \leq b, x \geq 0\}$$

let $P = \{x \in \mathbb{R}_+^n : Ax \leq b\}$ and $X = P \cap \mathbb{Z}^n$; then

$$z_{\text{IP}} = \max_x \{c^\top x : x \in X\} \leq z_{\text{LPR}} = \max_x \{c^\top x : x \in P\}$$

Def. a polyhedron $P = \{x \in \mathbb{R}_+^n : Ax \leq b\}$ is a **formulation** for X iff $X = P \cap \mathbb{Z}^n$

X = the feasible integer points — P must contain *exactly* X , not more, not less

there can be many **alternative formulations** P for X

example (Knapsack problem with three items):

$$X = \{(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)\} \subset \mathbb{Z}^3$$

$$\begin{array}{l|l|l} P_1 = \{x \in \mathbb{R}_+^3 : A^1 x \leq b^1\} & 3x + 3y + 3z \leq 5 & A^1 = \begin{bmatrix} 3 & 3 & 3 \end{bmatrix}, \quad b^1 = 5 \\ P_2 = \{x \in \mathbb{R}_+^3 : A^2 x \leq b^2\} & \begin{array}{l} x + y \leq 1 \\ x + z \leq 1 \\ y + z \leq 1 \end{array} & A^2 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad b^2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \end{array}$$

both contain exactly X : $P_1 \cap \mathbb{Z}^3 = P_2 \cap \mathbb{Z}^3 = X$ — **alternative formulations**; the polyhedra P_1, P_2 themselves are *different*

different formulations P_1 and P_2 may lead to different upper-bound quality

let $c = (1, 1, 1)$; hence $x_{\text{IP}}^* = (1, 0, 0)$ (or any unit vector) and $z_{\text{IP}} = c^\top x_{\text{IP}}^* = 1$

over P_1 : $x_{\text{LPR}}^* = (5/3, 0, 0)$, $z_{\text{LPR}}^1 = 5/3$

over P_2 : $x_{\text{LPR}}^* = (1/2, 1/2, 1/2)$, $z_{\text{LPR}}^2 = 3/2$

$$z_{\text{IP}} = 1 < z_{\text{LPR}}^2 = 3/2 < z_{\text{LPR}}^1 = 5/3$$

P_2 gives the tighter bound

takeaway this is why modelling is critical in IP: one X , many P — and you chose which one when you wrote the model down.

2. Which formulation is better?

Def. (Wolsey, Def. 1.5) given $X \subseteq \mathbb{Z}^n$ and formulations P_1, P_2 for X : P_1 is a **better** formulation than P_2 iff $P_1 \subset P_2$

both contain the same integer points; $P_1 \subset P_2$ = less *fractional* junk = an LP relaxation closer to the truth

how to prove $P_1 \subset P_2$ — $n \leq 3$: draw it. in general, do the following

$$\begin{array}{l|l} \text{step 1} & \text{show every } x \in P_1 \text{ is in } P_2 \quad (P_1 \subseteq P_2) \\ \text{step 2} & \text{exhibit one } \bar{x} \in P_2 \text{ with } \bar{x} \notin P_1 \quad (\text{strict}) \end{array}$$

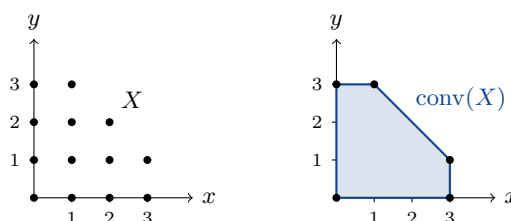
if $P_1 = P_2$: the two formulations are **equivalent**

if $P_1 \setminus P_2$ and $P_2 \setminus P_1$ are both nonempty: the two formulations are **incomparable**

takeaway trade-off: a tighter P usually costs more constraints.

3. $\text{conv}(X)$ — the ideal formulation, and the catch

Def. the **convex hull** of $X \subseteq \mathbb{R}^n$ is the minimal convex set containing X , denoted $\text{conv}(X)$



facts, stated without proof:

- 1 | $\text{conv}(X)$ is a polyhedron: $\text{conv}(X) = \{x \in \mathbb{R}_+^n : A^*x \leq b^*\}$ — so (A^*, b^*) is a formulation for X : the **ideal** one
- 2 | $\text{conv}(X) \subseteq P$ for *every* formulation P of X — nothing is better
- 3 | $\text{conv}(X)$ has **integer extreme points**

ideal reduction of IP to LP: one problem, $\max\{c^\top x : x \in X\}$, and two ways to write it down — the IP, and the **convex-hull problem** (CHP), an LP over $\text{conv}(X)$:

$$z_{\text{IP}} = \max_x \{c^\top x : x \in X\} \text{ an IP, } \quad z_{\text{CHP}} = \max_x \{c^\top x : x \in \text{conv}(X)\} \text{ just an LP}$$

Prop. $z_{\text{IP}} = z_{\text{CHP}}$, and an extreme-point optimum of the CHP is an optimum of the IP

Proof sketch. a CHP extreme-point optimum x_{CHP}^* is integer (fact 3) $\Rightarrow x_{\text{CHP}}^* \in X$: IP-feasible, so $z_{\text{CHP}} \leq z_{\text{IP}}$; and $X \subset \text{conv}(X)$ makes every IP point CHP-feasible, so $z_{\text{IP}} \leq z_{\text{CHP}}$ $\square$

so **you can ignore the integrality constraint**; for every other formulation ($P \supsetneq \text{conv}(X)$) we must impose integrality and solve an IP

example (the X drawn above):

$$\begin{aligned} \max \quad & 4x + 5y \\ \text{s.t.} \quad & 3x + y \leq 10, \\ & x + 3y \leq 10, \\ & x, y \geq 0 \text{ and int} \end{aligned} \quad \text{conv}(X) = \{x \geq 0, y \geq 0, x \leq 3, y \leq 3, x + y \leq 4\}$$

the vertices $(0,0), (3,0), (3,1), (1,3), (0,3)$ are all integer, and the CHP returns $(1,3)$, $z_{\text{CHP}} = 19 = z_{\text{IP}}$ — an integer answer with no integrality constraint anywhere

the CHP is *not* the LP relaxation: the relaxation optimizes over $P \supseteq \text{conv}(X)$ and overshoots; the CHP optimizes over $\text{conv}(X)$ and is exact

so why is this not the end of the course?

1. hard to characterise $\text{conv}(X)$: it may need an **exponential number** of rows
2. even harder: *proving* that what you have is $\text{conv}(X)$

takeaway $\text{conv}(X)$ would turn every IP into an LP. we almost never have it. so we do the next best thing: get P as tight as we can afford (§2), then pay for the rest with search — **branch & bound**, next.

4. Divide and conquer

two bounds squeeze z_{IP} from both sides:

$$\underbrace{c^\top \hat{x}}_{\hat{x} \in X: \text{ a feasible plan}} \leq z_{\text{IP}} \leq \underbrace{z_{\text{LPR}}}_{\text{a relaxation (§1)}}$$

the best feasible plan found so far is the **incumbent** $\hat{x}$, with value $\underline{z} = c^\top \hat{x}$ — the lower bound

Prop. if $X = X_1 \cup \dots \cup X_K$, then $z_{\text{IP}} = \max_j z^j$, where $z^j = \max\{c^\top x : x \in X_j\}$ — the best over X is the best of the pieces' bests

that is the whole idea: chop X into pieces; bound each piece with its LP relaxation; a piece whose bound cannot beat the incumbent is *thrown away without ever being searched* — **implicit enumeration**: every solution accounted for, none visited

5. Branch & bound

demo `demo6.branch.and.bound.py`

when can a piece X_k be thrown away? its LP relaxation tells you — three **prune** rules:

infeasibility	LP infeasible $\Rightarrow X_k = \emptyset$ (it sits inside the LP's feasible set)
bound	$z_{\text{LPR}} \leq \underline{z} \Rightarrow$ nothing in X_k beats the incumbent
optimality	x_{LPR}^* integer $\Rightarrow$ it is optimal for X_k — a new incumbent if it beats $\underline{z}$

notation. let $P_0 = \{x \in \mathbb{R}_+^n : Ax \leq b\}$, and similarly P_k for node k of the search tree:

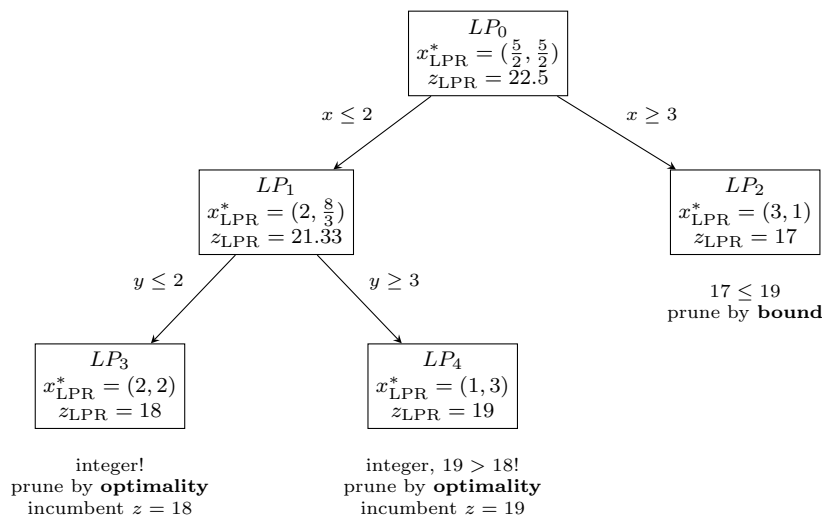
$$LP_0 : \max_x \{c^\top x : x \in P_0\}, \quad LP_k : \max_x \{c^\top x : x \in P_k\}$$

LP_0 is the LP relaxation of the IP; P_k is P_0 plus the bounds added by branching on the way down to node k , so LP_k relaxes the piece $X_k = P_k \cap \mathbb{Z}^n$

shorthand: $LP_k + \{x_j \leq a\} =$ the LP over $P_k \cap \{x : x_j \leq a\}$, e.g. $LP_1 = LP_0 + \{x \leq 2\}$; nodes are numbered in the order they are created

example (week 1's problem — the X drawn in §3):

$$\max 4x + 5y \quad \text{s.t.} \quad x + 3y \leq 10, \quad 3x + y \leq 10, \quad x, y \geq 0 \text{ and int}$$



root. solve LP_0 (drop integrality): $x_{LPR}^* = (\frac{5}{2}, \frac{5}{2})$, $z_{LPR} = 22.5$ — fractional $\Rightarrow$ **branch**

branch. $x \leq 2$ or $x \geq 3$: a decomposition (§4) that cuts out the fractional $x = 2.5$ and keeps every integer point

$$\begin{aligned}
 LP_1 &= LP_0 + \{x \leq 2\} & x_{LPR}^* &= (2, \frac{8}{3}), \quad z_{LPR} = 21.33 & \text{fractional} \Rightarrow \text{branch on } y \\
 LP_3 &= LP_1 + \{y \leq 2\} & x_{LPR}^* &= (2, 2), \quad z_{LPR} = 18 & \text{integer} \Rightarrow \text{prune by } \textit{optimality}; \text{ incumbent } \underline{z} = 18 \\
 LP_4 &= LP_1 + \{y \geq 3\} & x_{LPR}^* &= (1, 3), \quad z_{LPR} = 19 & \text{integer} \Rightarrow \text{prune by } \textit{optimality}; \text{ incumbent } \underline{z} = 19 \\
 LP_2 &= LP_0 + \{x \geq 3\} & x_{LPR}^* &= (3, 1), \quad z_{LPR} = 17 & 17 \leq 19 \Rightarrow \text{prune by } \textit{bound} \text{ (integer, but cannot win)}
 \end{aligned}$$

no active problems remain $\Rightarrow$ stop. $z_{IP} = 19$ at $(1, 3)$ — week 1's answer, now **proved**, with five LPs instead of enumerating

infeasibility? no piece of this tree is empty. in the demo, tilt the objective to $c = (10, 1)$: LP_0 moves to $(\frac{10}{3}, 0)$, and the branch $x \geq 4$ is empty ($3 \cdot 4 > 10$) — prune by infeasibility

Algorithm (Branch and bound)

- | | |
|---------------|---|
| 1. initialise | incumbent $\underline{z} = c^\top \hat{x}$ if a feasible $\hat{x}$ is known, else $\underline{z} = -\infty$; active list $L = \{LP_0\}$ |
| 2. stop? | if $L = \emptyset$: $\underline{z} = -\infty \Rightarrow$ report <i>infeasible</i> ; otherwise report $\hat{x}$ as optimal |
| 3. select | take a problem LP_k from L (and remove it); solve it for z_{LPR} and x_{LPR}^* |
| 4. prune | LP infeasible $\Rightarrow$ prune, go to 2 (by infeasibility)
$z_{LPR} \leq \underline{z} \Rightarrow$ prune, go to 2 (by bound)
x_{LPR}^* integer $\Rightarrow \hat{x} = x_{LPR}^*$, $\underline{z} = z_{LPR}$; prune, go to 2 (by optimality)
otherwise (x_{LPR}^* fractional) $\Rightarrow$ go to 5 |
| 5. branch | pick a fractional $x_j = v$ in x_{LPR}^* (the <i>branching variable</i> — there can be many to choose from); add two children LP_{m+1}, LP_{m+2} to L (LP_m = the last problem created), go to 2:
$P_{m+1} = P_k \cap \{x : x_j \leq \lfloor v \rfloor\}$, $P_{m+2} = P_k \cap \{x : x_j \geq \lceil v \rceil\}$ |

for **binary** IP branching is simpler: $x_j = 0$ or $x_j = 1$

every line is something you already had: step 3 is §1's $z_{IP} \leq z_{LPR}$; step 4 is the three prune rules above; step 5 is §4's decomposition

modern solvers do not only branch: they also add cuts at the nodes, tightening P as they go — **branch & cut**, week 15

takeaway branch & bound = divide (§4) + bound (§1) + throw away what cannot win (§5). three prune rules: optimality, bound, infeasibility. it never enumerates X ; it accounts for it.